CHAPTER 10 — LU Factorization

$$\underline{\underline{A}}\,\underline{x} = \underline{b}$$

- Gauss elimination becomes ineffizient when solving systems with same $\underline{\underline{A}}$ but different $\underline{b}$.
- LU factorization seperates the time consuming elimination step of the matrix A from the operations on the right hand side $\underline{b}$.

$$\underline{\underline{A}} = \underline{\underline{L}}\,\underline{\underline{U}}$$

- Once $\underline{\underline{A}}$ has been "factored" or "decomposed" → change $\underline{b}$ and solve effizietly.
- Gaussian elimination can be used for LU factorization.

Overview of LU factorization:

$$\underline{\underline{A}}\,\underline{x} = \underline{b} \qquad (1)$$

$$\underline{\underline{A}}\,\underline{x} - \underline{b} = 0$$

Assume (1) could be expressed as an upper triangular system:

$$\underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}}_{\underline{\underline{U}}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\underline{x}} = \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_{\underline{d}} \qquad (2) \Rightarrow \underbrace{\underline{\underline{U}}\,\underline{x} - \underline{d} = 0}$$

- Assume that there's a lower triangular matrix with 1's on the main diagonal:

$$\underline{\underline{L}} = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix} \quad (3)$$

If we multiply (2) by L, the result is (1)

$$L(Ux-d) = Ax-b \quad (4)$$

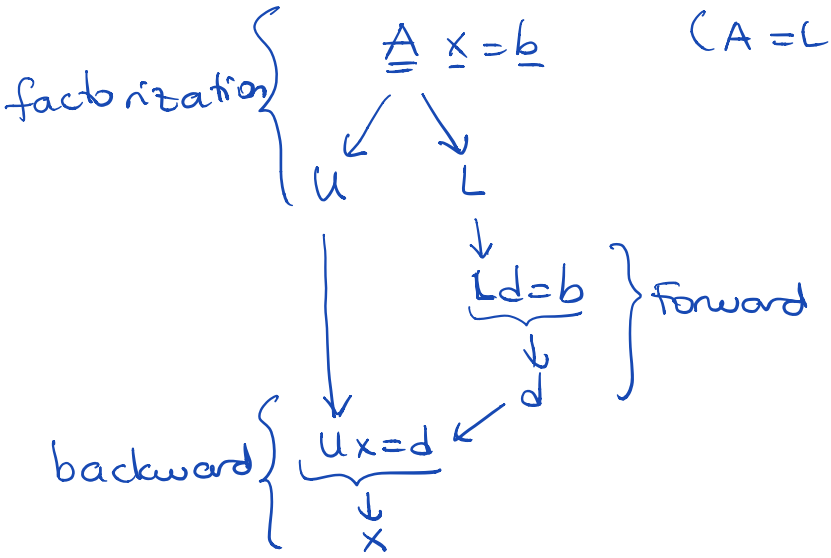
$$\Rightarrow \quad LU = A \quad (5)$$

and

$$Ld = b \quad (6)$$

Two-step method:

1) LU factorization of A
2) Substitution step: L and U are used to find solution for x for b
   - → use (6) ($\underline{\underline{L}}\,\underline{d} = \underline{b}$) to solve for $\underline{d}$ using forward substitution.
   - → Then use (2) ($\underline{\underline{U}}\,\underline{x} = \underline{d}$) to solve for $\underline{x}$ using back substitution.



Gauss Elimination as LU factorization

- " " can be used to decompose $A = LU$
- Forward elim. reduces A to an U matrix:

$$U = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix} \hookrightarrow \text{desired U format}$$

- L is also produced:

Consider: $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

First step:

- Multiply row 1 by: $f_{21} = \frac{a_{21}}{a_{11}}$
- Subtract from row 2 to eliminate a_{21}

- Multiply row 1 by: $f_{31} = \frac{a_{31}}{a_{11}}$
- Subtract from row 3 to eliminate a_{31}

- multiply the modified row 2 by $f_{32} = \frac{a'_{32}}{a'_{22}}$
- Subtract from 3rd row to eliminate a'_{32}

- Store the factors in another matrix:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix}$$

$A = LU$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

L $\qquad$ U

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ f_{21}a_{11} & f_{21}a_{12}+a'_{22} & f_{21}a_{13}+a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad f_{21} = \frac{a_{21}}{a_{11}}$$

a_{21}, a_{22}, a_{23}; $a_{33} \rightarrow f_{31}a_{13}+f_{32}a'_{23}+a''_{33}$

$f_{31}\cdot a_{11}$ $\qquad$ $f_{31}a_{12}+f_{32}a'_{22}$

- After LU decomp. solve for x as follows:
  - → Solve for d from $Ld=b$
  - → " " x from $Ux=d$

Ex Use Naive Gauss elim. to factor the system into LU decomposition:

$$7x_1 + 2x_2 - 3x_3 = -12$$
$$2x_1 + 5x_2 - 3x_3 = -20$$
$$x_1 - x_2 - 6x_3 = -26$$

$$A = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{7} & 1 & 0 \\ \frac{1}{7} & -0.29 & 1 \end{bmatrix} \begin{bmatrix} 7 & 2 & -3 \\ 0 & 4.42 & -2.14 \\ 0 & 0 & -6.19 \end{bmatrix}$$

f_{21}, f_{31}, f_{32}

L $\qquad$ U

LU factorization with Pivoting

Use permutation matrix:

1) Elimination → generated with elim. with partial pivoting

$$PA = LU$$

↳ store multiplier factors

↳ permutation matrix

2) Forward substitution:

$$PAx = LUx = Pb$$

$$Ld = Pb \Rightarrow \text{Solve for } d$$

3) Back substitution:

$$Ux = d \rightsquigarrow \text{Solve for } x$$

- LU factorization requires the same total # flops as Gauss elimination.

MATLAB: `[L,U] = lu(X)` → Built-in MATLAB function for LU decomp.

To Solve $Ax = b$

```
>> [L,U] = lu(A);
>>  d = L\b;
>>  x = U\d;
```

CHAPTER 11 – Matrix Inverse and Condition

$$A A^{-1} = A^{-1} A = I \quad , \quad A: \text{square matrix}$$

Calculating the inverse

- Compute in a column-by-column way:

if $b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ ⇝ solution to $Ax = b$ will be the first column of A^{-1}.

$x = A^{-1} b$ ↳ $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

if $b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ → sol^n to $Ax = b$ is $x = A^{-1} b \Rightarrow 2^{nd}$ column of A^{-1}.

- Can use LU factorization to solve for A^{-1}:

1st column of A^{-1} : $\underbrace{LU}_{d}\, x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow Ld = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{\text{solve}} d \Rightarrow Ux = d \xrightarrow{\text{Solve}} x$

2nd col. of A^{-1} : $\underbrace{LU}_{d}\, x = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow Ld = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \rightarrow d \Rightarrow Ux = d \rightarrow \underset{x}{\text{Solve}}$

3rd col. of A^{-1} : $LUx = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow Ld = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow d \Rightarrow Ux = d \rightarrow \underset{x}{\text{Solve}}$